

Gait Generation With Smooth Transition Using CPG-Based Locomotion Control for Hexapod Walking Robot

Haitao Yu, *Member, IEEE*, Haibo Gao, Liang Ding, *Member, IEEE*, Mantian Li, Zongquan Deng, and Guangjun Liu, *Senior Member, IEEE*

Abstract—This paper presents a locomotion control method based on central pattern generator (CPG) for hexapod walking robot to achieve gait generation with smooth transition. By deriving an analytical limit cycle approximation of the Van der Pol oscillator, a simple diffusive coupling scheme is proposed to construct a ring-shape CPG network with phase-locked behavior. The stability of the proposed network is proved using synchronization analysis with guaranteed uniform ultimate boundedness of the synchronous errors. In contrast to conventional numerical methods in tuning parameters of the CPG network, our method provides an explicit result that could analytically determine the network parameters to yield the rhythmic waveforms with prescribed frequency, amplitude, and phase relationship among neurons. Employing the proposed network to govern the swing/stance phase according to the profile of the resulting CPG signals, a locomotion control strategy for the hexapod robot is further developed to manipulate the leg movements during the gait cycle. By simply adjusting the phase lags of the CPG network, the proposed control strategy is capable of generating stable walking gaits for hexapod robots and achieving smooth transition among the generated gaits. The simulation and experimental results have demonstrated the effectiveness of the proposed locomotion control method.

Index Terms—Central pattern generator (CPG), gait generation, hexapod robot, legged locomotion, smooth transition, synchronization.

I. INTRODUCTION

LEGGED locomotion has been increasingly attracting attention among scientists and engineers over the last two decades [1]. The dexterous and agile performance of legged animals presents enormous inspiration in the design and control

of man-made mobile devices [2]–[4]. However, understanding the fundamental principle of how animals manage their own organs and limbs to deal with the coordination in multi-degree of freedom and multiaxial movements yet remains an open topic for biology and neurophysiology studies.

Recent neuroscience findings suggest that the central pattern generator (CPG) located in the spinal cord governs rhythmic limb activities without sensory feedback or regulation command from the brain-stem level [5]. Since the first practical model proposed by Matsuoka in 1987 [6], the concept of CPG has been successfully utilized in the locomotion control, especially in gait generation for diverse bioinspired robots such as snake/fish-like robot [7], humanoid robot [8], and quadruped robot [9]. By traditional approaches [10], [11], CPG is modeled as a group of neural oscillators formulated by coupled nonlinear differential equations. By carefully choosing the coupling scheme of the network, the population of coupled nonlinear oscillators could produce serials of stable quasi-periodic waveforms with specific frequency, amplitude, and phase relationship [12]. A challenge of investigating CPG system is to design the nonlinear oscillator model together with the coupling terms so that the resulting outputs satisfy the requirements of gait generation. Owing to the intrinsic nonlinearity of the neuron model as well as the strong coupling characteristics of the network, the parametrical investigations on such dynamical system have been traditionally implemented by the numerical methods such as evolutionary searching, empirical debugging, and other learning approaches [13]–[16]. These numerical methods inevitably isolate the parametric connection between the inherent dynamic behavior of the neuron model and the resulting waveform patterns. This weakness restricts the online adjustment ability of CPG network when applied in various circumstances.

Aside from the CPG-based gait generation, the issue of gait transition (a.k.a. gait switching) is of substantial importance for legged locomotion when encountering variation of traversing speed or terrain profile, as observed in biology experiments on horse and stick insect [17], [18]. Theoretical studies on this issue focus on the modeling the dynamic process by CMAC learning [19], cellular decomposition [20], gradient system [21], etc. Applying CPG concept to achieve gait transition has not been intensively investigated. Several continuous approaches accomplish the transition task by virtue of adjusting the topology structure [22], the coupling scheme [23], the model parameters of neuron [24] as well as resetting phase lags among neurons [8]. Another study on discrete process [25] accomplishes the task via

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H. Yu, H. Gao, L. Ding, M. Li, and Z. Deng are with the State Key Laboratory of Robotics and Systems, Harbin Institute of Technology, Harbin 150001, China (e-mail: yht@hit.edu.cn; gaohaibo@hit.edu.cn; liangding@hit.edu.cn; limt@hit.edu.cn; dengzq@hit.edu.cn).

G. Liu is with the Department of Aerospace Engineering, Ryerson University, Toronto, ON M5B 2K3, Canada (e-mail: gjliu@ryerson.ca).

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introducing the max-plus algebra. The gait transition is typically a transient process [8], [25] other than the steady state reported in conventional literatures [10], [11]. Such transient behavior can be theoretically recast as the issue of phase shifting of the coupled CPG network. In [26], a CPG model incorporating with an activation function to adjust the phase relation in cascaded oscillators is implemented on a snake-like robot to achieve gait transition. In [27], the gait transition for a turtle-like underwater robot is attained by adjusting the relative phases of a Hopf-based CPG network. The main challenge emerges in twofold: 1) the phase shifting should be steerable for the coupled network, which implies that not only the overall process should be stable, but also the desired phase lag and transition rate among the involved neurons could also be manipulated to suffice for locomotory applications [28]; 2) the smoothness during transition should be guaranteed because any abrupt or jerky signal produced by CPG unit might cause instability even *damage*.

This paper presents a novel CPG-based locomotion control method for hexapod walking robot to achieve gait generation with smooth transition. The CPG network is modeled as a group of coupled Van der Pol (VDP) oscillators. An analytical limit cycle approximation of the VDP is derived to analyze the phase relation between the adjacent neurons. A simple diffusive coupling scheme is then proposed to construct a ring-shape CPG network with phase-locked behavior. To generate stable gaits with smooth transition for hexapod robots, a corresponding locomotion control strategy is developed by integrating the proposed CPG network as the gait generator. The typical gait generation (i.e., tripod, tetrapod, and wave) with smooth transition for hexapodal walking is achieved by online adjusting the six phase lags of the CPG network. The effectiveness of the proposed method is demonstrated by simulation with a virtual hexapod model and by experiment with a physical robot platform.

The reminder of this paper is structured as follows. Section II describes the model of the coupled CPG network. The synchronization of the proposed CPG network is investigated in Section III. In Section IV, the CPG-based locomotion control strategy for hexapod walking robot is developed. Section V shows the simulation and experimental results. This paper ends with conclusion in Section VI.

II. CPG NETWORK

A. VDP Oscillator

The VDP oscillator used as the elementary neuron model of the CPG network is formulated by

$$\ddot{x} + \varepsilon (bx^2 - 1) \dot{x} + \omega_0^2 x = 0 \quad (1)$$

where x is the state variable of the nonlinear system; ε is a scalar coefficient quantifying the strength of the nonlinear damping ($0 < \varepsilon \ll 1$); b is the amplitude adjustment parameter ($b > 0$), and ω_0 is the natural frequency of the auxiliary linear system (when $\varepsilon = 0$). The state-space format of (1) is given by

$$\dot{z} = \mathbf{F}(z) = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} \omega_0 y \\ -\omega_0 x \end{bmatrix} + \varepsilon \begin{bmatrix} x(1 - bx^2/3) \\ 0 \end{bmatrix} \quad (2)$$

where $z = [x, y]^T$ denotes the state variable of VDP. The accurate solution of VDP is inaccessible due to the nonintegrability of the damping terms in (1). As an alternative, the method of multiple scales (MMS) in perturbation theory is introduced to obtain the approximate analytical solution. Let ξ and η be the new time scales to yield

$$\xi = \varepsilon t, \eta = \left(1 + \sum_{i=2}^N \varepsilon^i \omega_i\right) t \quad (3)$$

where ω_i is the undetermined frequency; $N \in \mathbb{Z}^+$ is a positive constant that represents the highest order of ε in the new time scale η . Herein, ε is regarded as a perturbation parameter such that the approximate solution of (1) can be expressed as

$$x = x_0(\xi, \eta) + \varepsilon x_1(\xi, \eta) + \varepsilon^2 x_2(\xi, \eta) + \dots \quad (4)$$

where x_i is an approximation to be determined. Substituting (4) into (1) to expand the left-hand side of the resulting equation in power series of ε , we have

$$\varepsilon^0 : \frac{\partial^2 x_0}{\partial \eta^2} + \omega_0^2 x_0 = 0 \quad (5)$$

$$\varepsilon^1 : \frac{\partial^2 x_1}{\partial \eta^2} + \omega_0^2 x_1 = -2 \frac{\partial^2 x_0}{\partial \xi \partial \eta} + (1 - bx_0^2) \frac{\partial x_0}{\partial \eta}. \quad (6)$$

Here, the approximation is restricted within the first-order expansion of ε to ease the algebra complexity of the resulting expression. Following the standard MMS procedure, we finally obtain the limit cycle approximation of VDP as

$$x \approx \frac{2}{\sqrt{b + (4/a_0^2 - b) e^{-\varepsilon t}}} \cos(\omega_0 t + \varphi_0) \quad (7)$$

where a_0 and φ_0 are determined by the initial condition of (1). The steady-state solution can be obtained via $t \rightarrow \infty$ as

$$x_\infty(t) = \frac{2}{\sqrt{b}} \cos(\omega_0 t + \varphi_0), y_\infty(t) = -\frac{2}{\sqrt{b}} \sin(\omega_0 t + \varphi_0). \quad (8)$$

It is obvious from (7) and (8) that each model parameter in (1) plays an irreplaceable role in dominating the limit cycle behavior of VDP. The amplitude and frequency of the limit cycle can be independently tuned by b and ω_0 , respectively. The coefficient ε shapes the output profile of VDP. The initial phase φ_0 , a key parameter to quantify the phase lag between the connected neurons, will be determined in Section IV.

B. Diffusive Coupling Scheme

The topology of the CPG network is determined by the coupling term of the connected neurons. For simplicity, all the neurons of the network $\{\text{CPG}_i\}$ produce the periodic signals with identical amplitude and frequency. The waveforms of the CPG network is solely determined by phase lags φ_i among the connected neurons. The constant phase lag can be defined between the CPG_i and CPG_j when attaining the steady state as

$$\theta_i := \varphi_i, 0 - \varphi_j, 0 \quad (9)$$

where θ_i is the desired phase lag between the adjacent neurons CPG_i and CPG_j when the whole network attains the steady

state; $\varphi_{i,0}$ and $\varphi_{j,0}$ are the initial phase of the two neurons involved. Then, according to the derived limit cycle expression in (8), the following relation holds:

$$z_{j,\infty}(t) = \mathbf{R}(\theta_i) z_{i,\infty}(t) \quad (10)$$

where $\mathbf{R}(\theta_i)$ is a planar rotation matrix and satisfies

$$\mathbf{R}(\theta_i) = \begin{bmatrix} \cos \theta_i & \sin \theta_i \\ -\sin \theta_i & \cos \theta_i \end{bmatrix}. \quad (11)$$

Specifically, the steady states of CPG_i and CPG_j are given by

$$z_{i,\infty}(t) = \left[\frac{2}{\sqrt{b}} \cos(\omega_0 t + \varphi_{i,0}), -\frac{2}{\sqrt{b}} \sin(\omega_0 t + \varphi_{i,0}) \right]^T$$

$$z_{j,\infty}(t) = \left[\frac{2}{\sqrt{b}} \cos(\omega_0 t + \varphi_{j,0}), -\frac{2}{\sqrt{b}} \sin(\omega_0 t + \varphi_{j,0}) \right]^T.$$

Equation (10) describes the desired relationship between the two connected neurons. On this basis, the diffusive coupling scheme of the CPG network is formulated as

$$\dot{z}_i = \begin{cases} \mathbf{F}(z_i) + \gamma(\mathbf{R}(\theta_i) z_{i+1} - z_i), & \text{for } i = 1, 2, \dots, N-1 \\ \mathbf{F}(z_i) + \gamma(\mathbf{R}(\theta_i) z_1 - z_i), & \text{for } i = N \end{cases} \quad (12)$$

where z_i is the state vector of the i th neuron; γ is the constant coupling gain to be determined in accordance with the synchronization of the network in Section III. The rotation matrix $\mathbf{R}(\theta_i)$ is regarded as the coupling weight between the two adjacent neurons and satisfies

$$\prod_{i=1}^N \mathbf{R}(\theta_i) = \mathbf{I}_2. \quad (13)$$

For each neuron of the network, the dynamics of z_i in (12) can be divided into two terms: the original VDP term and the diffusive coupling term. The former inherits the limit cycle behavior of VDP preserving the output x_i with a bounded amplitude and constant frequency. The latter couples the current CPG with the anterior and posterior CPGs. Fig. 1 illustrates the topology of the CPG network with N nodes that forms a ring.

III. SYNCHRONIZATION OF THE CPG NETWORK

A. Synchronization Analysis of the CPG Network

In this section, we focus on the performance of the CPG network established in Section II. The synchronization analysis will be implemented with necessary theoretical preliminaries to investigate the global behavior of the proposed CPG network.

The solution of the isolate node in network (12) can be conveyed as

$$\dot{s} = \mathbf{F}(s). \quad (14)$$

Here, $s(t)$ denotes the steady state of VDP. We have the following properties.

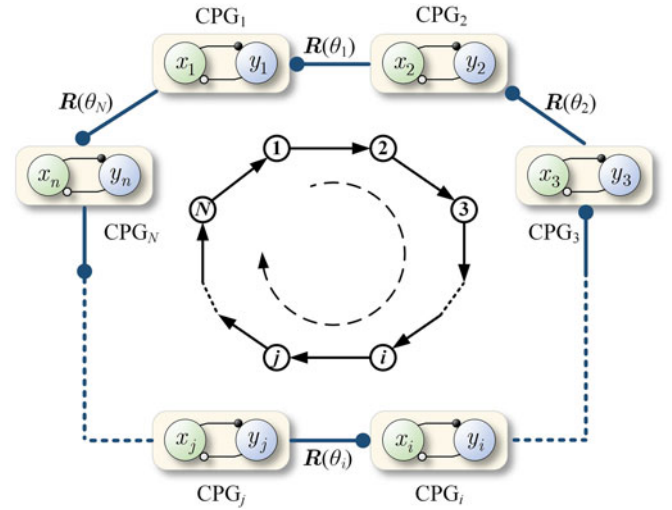


Fig. 1. Topology of the CPG network in ring. The topology of the network is considered as a directed graph with N nodes.

Property 1: For a given periodic trajectory $s(s_0, t_0 : t)$ of (14), there exists a compact set $\mathbf{B}_\delta \subset \mathbb{R}^2$ such that if $s_0 \in \mathbf{B}_\delta$, then $s(t) \in \mathbf{B}_\delta$ for $t > t_0$. Specifically $s(t)$ is bounded by

$$\|s(t)\| < \delta, \quad \delta \in \mathbb{R}^+ \quad (15)$$

where $\|\cdot\|$ denotes the Euclidean norm. According to the approximation of VDP in (8), we choose $\delta = 3/\sqrt{b}$ to quantify a subset of basin of attraction of VDP.

Property 2: Given a function $\mathbf{F}(z) : \mathbb{R}^2 \mapsto \mathbb{R}^2$ defined in (2) with $z = [x, y]^T$, if z is bounded by $\|z\| \leq \mathbb{R}^+$, then the following inequality holds for $\mathbf{F}$

$$\|\mathbf{F}(z)\| \leq (\omega_0 + \varepsilon) \delta + \frac{\varepsilon b}{3} \delta^3. \quad (16)$$

Proof: According to the definition of $\mathbf{F}$ in (2), we have

$$\|\mathbf{F}(z)\| \leq \omega_0 \|z\| + \varepsilon \left(\|z\| + \frac{b}{3} \|z\|^3 \right) \leq (\omega_0 + \varepsilon) \delta + \frac{\varepsilon b}{3} \delta^3.$$

Thus, the inequality (15) is proven directly. ■

Given the phase lags $\{\theta_i\}$ of the network, the synchronization objective of (12) is to guarantee that each node converges into its target synchronous state, namely, achieving $z_i \rightarrow s_i$ in a finite time for $i = 1, \dots, N$. Here, s_i is the target synchronous state of z_i for node i and given by

$$s_i = \Gamma_i^{-1} s$$

with

$$\Gamma_i = \begin{cases} \mathbf{I}_2, & \text{for } i = 1 \\ \prod_{j=1}^{i-1} \mathbf{R}(\theta_j), & \text{for } 1 < i \leq N \end{cases}.$$

To facilitate the subsequent analysis, the following coordinate transformation is introduced as

$$\tilde{z}_i = \Gamma_i z_i.$$

Then, the original CPG network can be rewritten into

$$\dot{\tilde{z}}_i = \Gamma_i \mathbf{F}(\Gamma_i^{-1} \tilde{z}_i) + \gamma (\tilde{z}_{i+1} - \tilde{z}_i), \text{ for } i = 1, 2, \dots, N. \quad (17)$$

Note that the subscript $i = N + 1$ denotes i . It is obvious that the overall nodes of the network (17) will synchronize if and only if the following condition is attained:

$$\tilde{z}_1 = \tilde{z}_2 = \tilde{z}_3 = \dots = \tilde{z}_n = s.$$

Then, $S = (s^T, s^T, \dots, s^T)^T$ is a synchronous solution of the network (17). The following lemmas are needed to derive the main result.

Lemma 1[29]: Given the vector function $F: \mathbb{R}^2 \times \mathbb{R}^+ \mapsto \mathbb{R}^2$ in (2) where b, ω_0 and ε are strictly positive constants, together with two arbitrary vectors $z_1, z_2 \in \mathbb{R}^2$ and planar rotation matrix $\mathbf{R}$ defined in (17), then the following inequality holds:

$$(z_1 - z_2)^T (\mathbf{R}\mathbf{F}(\mathbf{R}^{-1}z_1) - \mathbf{R}\mathbf{F}(\mathbf{R}^{-1}z_2)) \leq \varepsilon \|z_1 - z_2\|^2. \quad (18)$$

Lemma 2[30]: Let $\mathbf{L}$ denote the Laplacian matrix of an undirected connected network with node N , the corresponding eigenvalue satisfies $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_N$. For a given nonzero vector $z \in \mathbb{R}^{N \times 1}$, there exists $\xi \in (0, \lambda_N)$ such that

$$\xi = \frac{v^T \mathbf{L} v}{v^T v}. \quad (19)$$

Lemma 3: Given a differentiable scalar function $V(t) \geq 0$ defined for $t \geq 0$, if the following holds:

$$\dot{V}(t) < -\rho V(t) + \mu \sqrt{V(t)} \quad (20)$$

where ρ and μ are both positive constants, then

$$\sqrt{V(t)} < \frac{\mu}{\rho} - \left(\frac{\mu}{\rho} - \sqrt{V(0)} \right) e^{-\frac{\rho t}{2}}. \quad (21)$$

The proof is omitted here for brevity.

Based on the aforementioned lemmas, the following result is now ready to be presented.

Theorem 1: Given a CPG network with N nodes formulated by (17), define the synchronous error for i th node as

$$e_i := \tilde{z}_i - s, \quad i = 1, 2, \dots, N \quad (22)$$

with $e := (e_1^T, e_2^T, \dots, e_N^T)^T$, then for a suitable design of the coupling gain γ , the CPG network (17) will be globally synchronized and the synchronous error $e(t)$ are uniformly ultimately bounded (UUB). Additionally, $e(t)$ can be kept as arbitrary small as desired by increasing the coupling gain γ .

Proof: Consider a Lyapunov function candidate as follows:

$$V(t) := \frac{1}{2} \sum_{i=1}^N e_i^T e_i. \quad (23)$$

Differentiating $V(t)$ along the node dynamics (17) and adding and subtracting $\Gamma_i \mathbf{F}(\Gamma_i^{-1} s)$, we have

$$\dot{V}(t) = \sum_{i=1}^N e_i^T (\dot{\tilde{z}}_i - \dot{s})$$

$$\begin{aligned} &= \underbrace{\sum_{i=1}^N e_i^T (\Gamma_i \mathbf{F}(\Gamma_i^{-1} \tilde{z}_i) - \Gamma_i \mathbf{F}(\Gamma_i^{-1} s))}_{\dot{V}_1(t)} \\ &\quad + \underbrace{\sum_{i=1}^N e_i^T (\Gamma_i \mathbf{F}(\Gamma_i^{-1} s) - \mathbf{F}(s))}_{\dot{V}_2(t)} + \underbrace{\gamma \sum_{i=1}^N e_i^T (\tilde{z}_{i+1} - \tilde{z}_i)}_{\dot{V}_3(t)}. \end{aligned} \quad (24)$$

Using Lemma 1 with $z_1 = \tilde{z}_i$, $z_2 = s$ and $\mathbf{R} = \Gamma_i$ to yield

$$\begin{aligned} \dot{V}_1(t) &= \sum_{i=1}^N (\tilde{z}_i - s)^T (\Gamma_i \mathbf{F}(\Gamma_i^{-1} \tilde{z}_i) - \Gamma_i \mathbf{F}(\Gamma_i^{-1} s)) \\ &\leq \sum_{i=1}^N \varepsilon \|e_i\|^2 = \varepsilon e^T e. \end{aligned} \quad (25)$$

Thus, $\dot{V}_1(t)$ is bounded. Concentrating on $\dot{V}_2(t)$ and using the Cauchy–Schwarz inequality to obtain

$$\dot{V}_2(t) \leq \sum_{i=1}^N \|e_i^T\| (\|\mathbf{F}(\Gamma_i^{-1} s)\| + \|\mathbf{F}(s)\|).$$

Applying property 1 together with (15), $\dot{V}_2(t)$ can be further estimated via (16) as

$$\dot{V}_2(t) < 2M(\delta) \sum_{i=1}^N \|e_i\| \leq 2M(\delta) \sqrt{N} e^T e \quad (26)$$

where the scalar function $M(\delta)$ is given by

$$M(\delta) := (\omega_0 + \varepsilon) \delta + \frac{\varepsilon b}{3} \delta^3.$$

Therefore, $\dot{V}_2(t)$ is also bounded. The remaining term $\dot{V}_3(t)$ can be simplified by substituting (22) into $\dot{V}_3(t)$ to yield

$$\dot{V}_3(t) = \gamma \sum_{i=1}^N e_i^T (e_{i+1} - e_i) = -\frac{\gamma}{2} e^T ((P + P^T) \otimes I_2) e$$

where $\otimes$ denotes the Kronecker product and the incidence matrix $P = (p_{ij})_{N \times N}$ of the directed graph (abstracted in Fig. 1) evolved from the network (17) is defined as follows: if node i points to node j with an oriented edge, then $p_{ij} = -1$; if node i leaves node j with an oriented edge, then $p_{ij} = 1$; otherwise $p_{ij} = 0$. Note that $P + P^T$ is equivalent to the Laplacian matrix $\mathbf{L}$ of the network (17), namely $P + P^T = \mathbf{L}$ holds. Consequently we have

$$\dot{V}_3(t) = -\frac{\gamma}{2} e^T (\mathbf{L} \otimes I_2) e.$$

Using Lemma 2, there exists $\xi \in (0, \lambda_N)$ such that

$$\dot{V}_3(t) = -\gamma \xi e^T e. \quad (27)$$

Up to this point, combining (25)–(27), we obtain

$$\dot{V}(t) < -C_\rho V + C_\mu \sqrt{V}$$

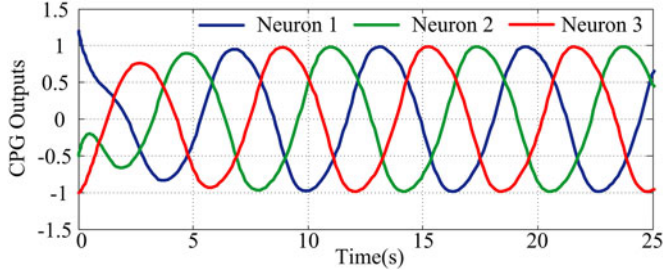


Fig. 2. Evolution of the CPG network with structure parameters $b = 4$, $e = 0.3$, $\omega_0 = 1$, and $\gamma = 1$.

where

$$C_\rho = 2(\gamma\xi - \varepsilon), \quad C_\mu = 2M(\delta)\sqrt{2N}.$$

And C_ρ is guaranteed positive as long as $\gamma > \varepsilon/\xi$. By using Lemma 3 with replacing ρ and μ with C_ρ and C_μ , respectively, it is obvious that V is negative outside a compact set Θ_e as

$$\Theta_e := \left\{ e(t) \mid V(t) = \frac{1}{2} e(t)^T e(t) < \frac{C_\mu^2}{C_\rho^2} := V_\infty \right\}.$$

Therefore, Θ_e is an invariant set such that if $V(0) < V_\infty$, then $V(t) < V_\infty$ for all $t > 0$, which implies that all the synchronous errors in the closed-loop system are UUB in Θ_e . Additionally, by increasing the coupling gain γ , the compact set Θ_e will be kept arbitrarily small according to (21). The synchronous error e_i can be restricted arbitrarily small in accordance with practical requirements. The proof is completed. ■

B. Phase-Locked Behavior

The advantage of the synchronized CPG network is to exhibit a phase-locked behavior via a collection of the coupling nodes. To verify the effectiveness of the proposed coupling scheme, a coupled CPG network is established with three VDP oscillators in (12). The desired phase lag between the adjacent neurons is preset at $\theta_i = 4\pi/3$ for $i = 1, 2, 3$. The indicator to quantify the synchronous error is defined as

$$E(t) := \sqrt{\frac{1}{N} \sum_{i=1}^N (\mathbf{R}(\theta_i) z_{i+1} - z_i)^T (\mathbf{R}(\theta_i) z_{i+1} - z_i)}. \quad (28)$$

Fig. 2 shows the output signals of the CPG network. Evolving from a disordered state at $t = 0$ s, the network rapidly converges into a periodic waveform almost after $t = 5$ s and ultimately attains synchronization such that all nodes produce the identical limit cycle trajectory with the desire phase lag. It can be clearly observed in Fig. 3 that the amplitude of $E(t)$ decreases within 2.5s and maintains bounded with considerably small synchronous errors.

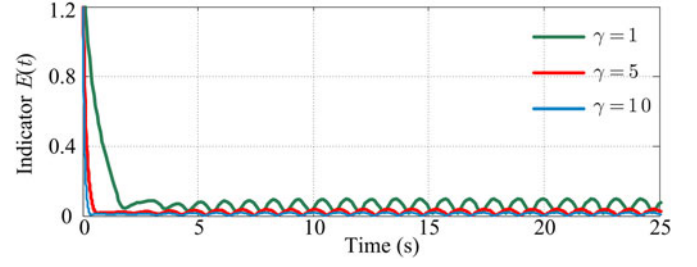


Fig. 3. Synchronous indicator $E(t)$ versus time with different coupling gains of the CPG network.

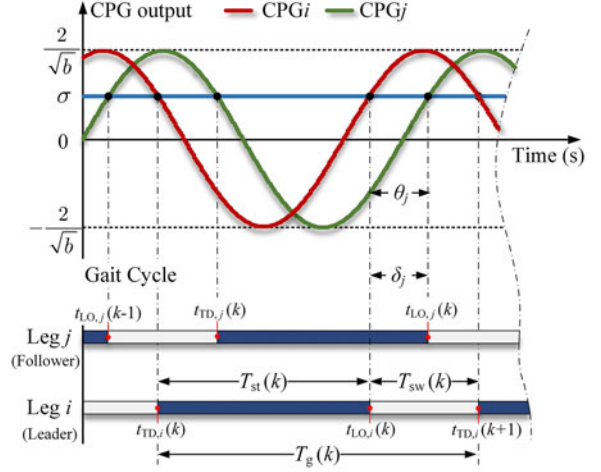


Fig. 4. Mapping from the CPG signals to the gait specifications. The subscript TD and LO represent the touchdown and liftoff event of the foot, respectively.

IV. CPG-BASED LOCOMOTION CONTROL STRATEGY

This section focuses on the development of the locomotion control strategy for hexapod robot. The proposed strategy integrates the CPG network as the gait generator via applying the rhythmic CPG outputs as the gait cyclic reference to allocate swing/stance subphase during walking. Subsequently, a foot trajectory planning is presented to manipulate limb movement, working in conjunction with a joint-space tracking controller to enforce the desired trajectory for each leg.

A. Mapping CPG Signals to Gait Specifications

The key gait specifications for legged locomotion are the duty factor (DF) and the phase sequence (PS). The DF β_{DF} specifies the duration proportion that the stance phase occupies in the entire gait cycle, and can be quantified by

$$\beta_{DF} := \frac{T_{st}}{T_g} = \frac{T_{st}}{T_{st} + T_{sw}}$$

where T_{st} and T_{sw} are the duration of the stance and swing phase, respectively, T_g is the gait cycle and satisfies $T_g = T_{st} + T_{sw}$.

The PS determines the swing-leg order by quantifying the time delay (measured in percentage during gait cycle) between the leader leg and the current swing leg (follower). Without loss of generality, Leg1 is assumed to be the leader of all six legs.

TABLE I
RELEVANT PARAMETERS OF TYPICAL GAIT FOR HEXAPOD LOCOMOTION

Gait	Gait Specifications		CPG Network Parameters	
	Duty Factor β_{DF}	Phase Sequence δ_i (Leg 1 ~ 6)*	Threshold σ	Phase Lag θ_i (CPG 1 ~ 6)
Tripod	$\frac{1}{2}$	0, 1/2, 0, 1/2, 0, 1/2	0	$\pi, \pi, \pi, \pi, \pi, \pi$
Tetrapod	$2/3$	0, 1/3, 2/3, 2/3, 0, 1/3	1/2	$2\pi/3, 2\pi/3, 0, 2\pi/3, 2\pi/3, 4\pi/3$
Wave	$5/6$	0, 2/6, 4/6, 1/6, 3/6, 5/6	$\sqrt{3}/2$	$2\pi/3, 2\pi/3, \pi, 2\pi/3, 2\pi/3, \pi/3$

*See Fig. 5 for the illustration of leg number.

The PS can be defined by

$$\delta_{PS} := (0, \delta_2, \delta_3, \dots, \delta_6)$$

where δ_i ($i = 2 \sim 6$) denotes the time delay of Leg i to Leg 1 and $\delta_1 = 0$ always holds for the leader leg.

Considering that the CPG unit produces quasi-sinusoidal signal with prescribed phase lag, we introduce a variable σ as a threshold to scale the CPG output. The mapping rules (as illustrated in Fig. 4) can be summarized as follows.

Rule I (Determining the DF): For i th CPG unit, the duration that $x_i \geq \sigma$ holds represents swing phase for legged locomotion, otherwise $x_i < \sigma$ for stance phase. The threshold σ is adjustable within the range $[-2/\sqrt{b}, 2/\sqrt{b}]$.

By using (8), the limit cycle can be normalized as a unit circle by setting $b = 4$. Then, the DF is determined via

$$\beta_{DF} \approx \frac{1}{2} + \frac{\arcsin \sigma}{\pi}, \quad \sigma \in [-1, 1]. \quad (29)$$

Rule II (Determining the PS): We employ the CPG network with $N = 6$ as the waveform signal generator for gait control. Each neuron solely governs the motion of a single leg for the hexapod robot. The phase lag between the adjacent neurons is mapped into the PS by

$$\delta_i = \begin{cases} 0, & \text{for } i = 1 \\ \frac{1}{2\pi} \left(\left(\sum_{j=1}^{i-1} \theta_j \right) \bmod 2\pi \right), & \text{for } i = 2 \sim 6 \end{cases}. \quad (30)$$

Note that the modulo-operator in (30) is utilized to normalize the accumulated sum of θ_i into $[0, 2\pi]$. The typical gait for hexapodal locomotion can be attained by appropriately setting the parameters of the CPG network as listed in Table I. We focus on the walking gait for the hexapod robot, therefore the DF β_{DF} in this paper is restricted within $[0.5, 1)$.

B. Locomotion Control Strategy for Hexapod Robot

The proposed locomotion control strategy for hexapod walking robot contains three parts: the desired motion regulation (DMR), the CPG-based gait generator as well as the joint-space tracking controller (as illustrated in Fig. 5). The strategy takes the desired walking speed v and a target gait G with the specifications $(\beta_{DF}, \delta_{1 \sim 6})$ as input. The robot is assumed to

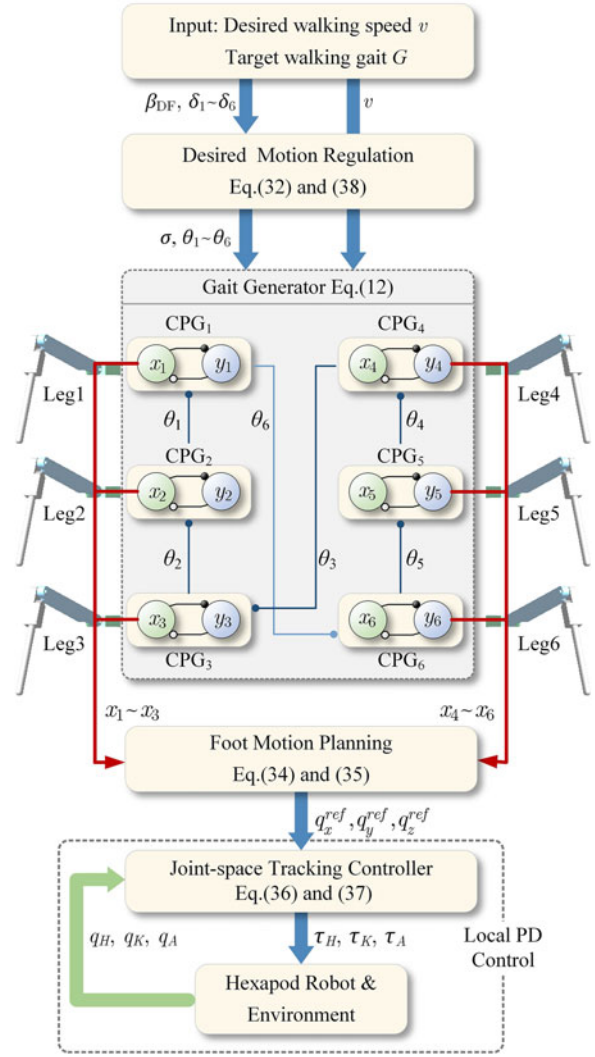


Fig. 5. Schematic of the CPG-based locomotion control strategy.

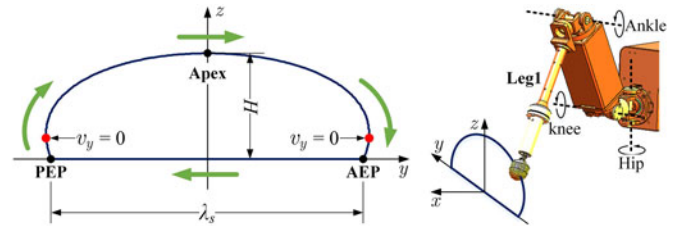


Fig. 6. Foot trajectory for a single leg of the hexapod robot during the gait cycle. Red solid circles denote the reversing point of horizontal velocity.

move forward at a constant velocity in stance phase, resulting the following relation:

$$v = \frac{\omega_0 (y_{AEP} - y_{PEP})}{2\pi\beta_{DF}} = \frac{\omega_0 \lambda_s}{\pi + 2 \arcsin \sigma} \quad (31)$$

where v is the forward velocity of the robot, y_{AEP} and y_{PEP} are the anterior and posterior extreme position of the foot on the ground, λ_s is the stride length (exemplified in Fig. 6).

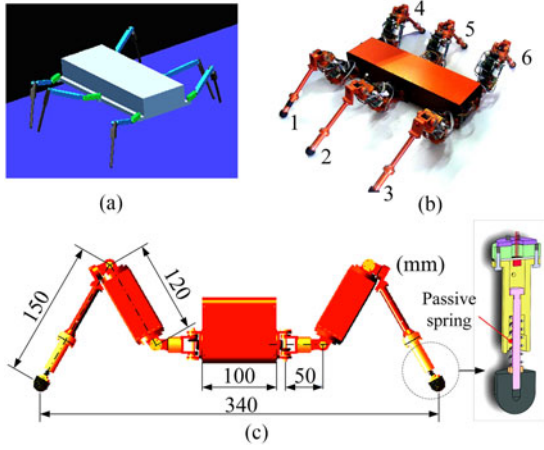


Fig. 8. Overview of the hexapod robot prototype. (a) Virtual simulation model, (b) hexapod robot platform, and (c) key dimensions with foot details.

TABLE II
RELEVANT MECHANICAL AND CONTROL PARAMETERS OF THE HEXAPOD ROBOT

Mechanical parameters		PD controller parameters	
Robot mass (total)	5.16 kg	K_P (hip joint)	45
Body mass	2.21 kg	K_D (hip joint)	2
Leg mass (each)	0.49 kg	K_P (Knee joint)	80
Body length	250 mm	K_D (Knee joint)	3.5
Body height	120 mm	K_P (Ankle joint)	10
Foot clearance	40 mm	K_D (Ankle joint)	0.25

C. Gait Specifications Adjustment for Transition

The concept *gait transition* addressed in this paper refers to the process that a robot switches its walking pattern from the current gait into another accessible gait and maintains the original direction of motion. Any sudden or sharp change of the network parameters and gait specifications will cause discontinuous even infeasible movement of leg joint, which is harmful to the joint actuator, and should be avoided during the gait transition. To this end, we propose an online adjustment for the gait specification pair (β_{DF}, δ_i) by introducing the sigmoid function as

$$\beta_{DF}(t) = \beta_{DF}^- + \frac{(\beta_{DF}^+ - \beta_{DF}^-)}{1 + e^{-\kappa(t-t_{ts})}}, \delta_i(t) = \delta_i^- + \frac{(\delta_i^+ - \delta_i^-)}{1 + e^{-\kappa(t-t_{ts})}} \quad (38)$$

where t_{ts} is the gait transition time, the superscript “+”, “−” distinguish the current and the subsequent state, and κ is a time constant to adjust the transition duration.

V. SIMULATION AND EXPERIMENTAL EVALUATION

A. Experiment Setup

The virtual model and physical hexapod robot used in simulation and experiment are shown in Fig. 8. The robot-environment information including joint angle/torque, body posture together with contact force are recorded simultaneously in simulation. Fig. 8(b) and (c) illustrates the dimensions of the hexapod platform developed for the experiment. Table II shows the relevant

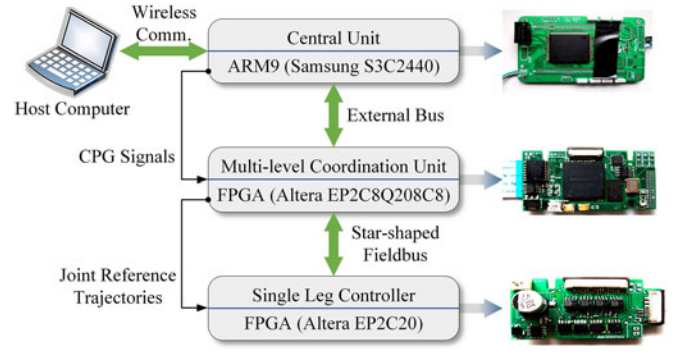


Fig. 9. Illustration of the hardware configuration.

mechanical and control parameters. Each leg with three revolution joints driven by a 15-W brushless DC motor (MAXON, EC32) with torque constant 49 mNm/A is located on the aluminum alloy body. A particular design on the end effector with passive spring structure is applied to reduce the foot-ground impact which is detected by a contact sensitive cell with binary signal mounted at the toe of each foot.

A three-layered hardware configuration for the hexapod walking robot is developed as shown in Fig. 9. The top layer employs an ARM9 microcontroller (Samsung, S3C2440) as the central processing unit to execute the computing of the CPG-based algorithm in C-programming via the fourth-order Runge–Kutta method. The motion command is transmitted from the host computer to the central unit via wireless communication. The middle layer takes charge of receiving the produced CPG signals from the top layer, transferring into the reference trajectories for each joint of individual leg as well as multitasking scheduling on one FPGA-based chip-on board (Altera, CycloneII series, EP2C8Q208C8) running at 200 Hz. A FPGA-based (Altera, CycloneII series, EP2C20) single leg controller integrated with H-bridge gate driver IC (FairChild, Mosfet-FDS4885) is developed for the servo motor on the bottom layer to indirectly implement the local PD control (37) by transferring the current commands into the driving torques for three revolution joints of each leg. This layer, connected to the middle layer via an extensible restrained star-shaped fieldbus, also executes sensor sampling task including the motor encoder, contact cell, Hall-effect transducer, and current sensor to supervise the internal and external states of the robot when interacting with environment.

B. Simulation Results

The simulation is performed to testify the effectiveness of the proposed strategy in achieving stable tripod, tetrapod, and wave gait generation with smooth transition for the virtual hexapod model. A flat surface is created in the virtual environment as shown in Fig. 8(a). A flat surface is created in the virtual environment as shown in Fig. 8(a). The foot-ground reaction forces in normal and tangential direction are, respectively, modeled as the linear damper-spring model with the environment stiffness coefficient $K_e = 10^6$ N/m and damping coefficient $B_e = 2.0 \times 10^3$ Ns/m, and the Coulomb friction with the static coefficient $\mu_s = 0.75$ and dynamic coefficient $\mu_d = 0.50$.

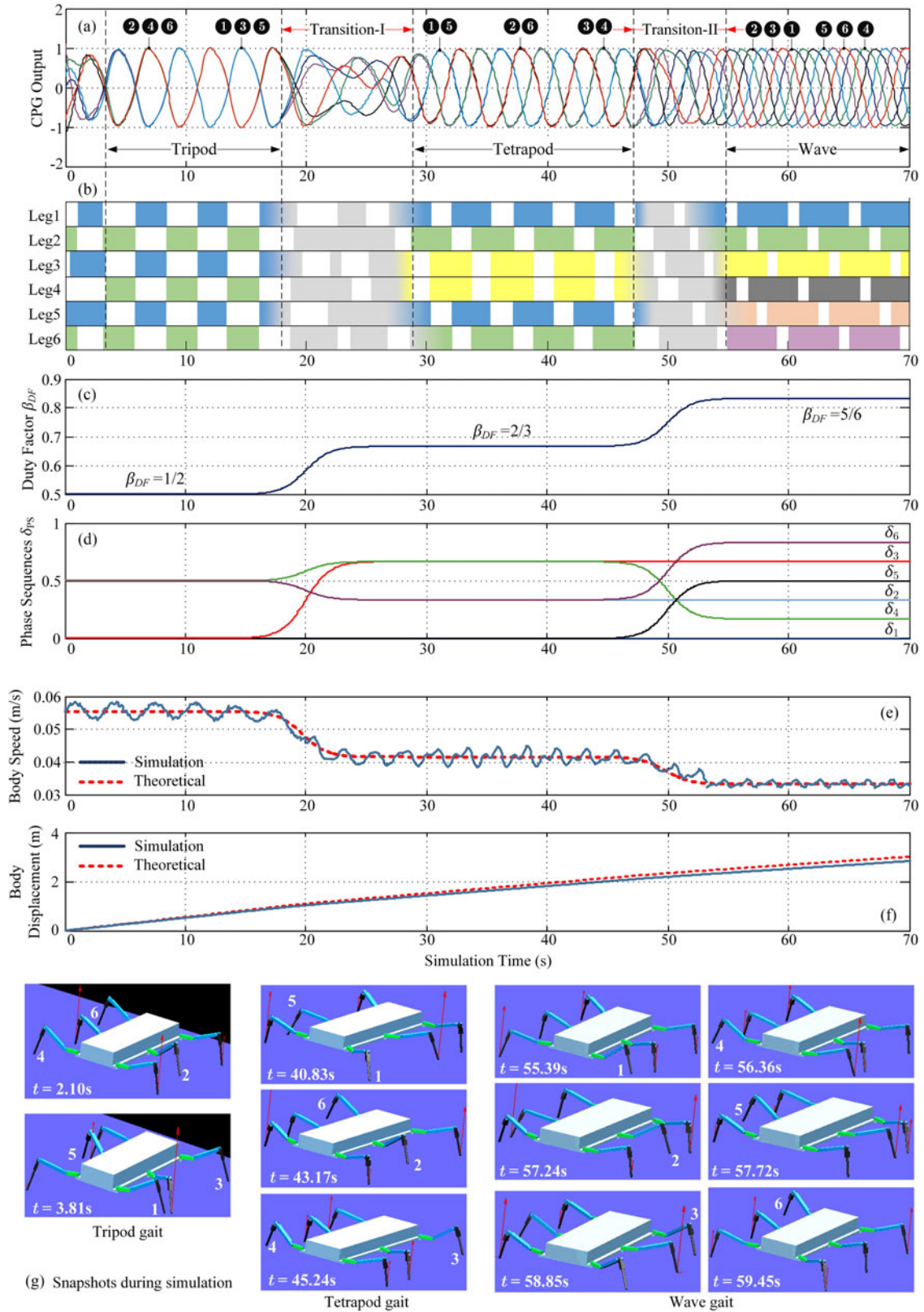


Fig. 10. Simulation results on a virtual hexapod robot. (a) Outputs of the coupled CPG network. The individual CPG unit is entitled with number in solid circle. (b) Resulting footprint diagram of the gait cycle during gait transition. The color/transparent bars represent legs in stance/swing phase. The gray areas indicate the transition process of the virtual robot. (c) Corresponding DF profile during simulation. (d) Corresponding PSs profile during simulation. (e) Body speed of the hexapod robot during simulation. (f) Body displacement of the hexapod robot during simulation. (g) Snapshots of the hexapod robot in simulation. The leg in swing phase is labeled with its leg number. The time instant in simulation is given at the bottom of each snapshot.

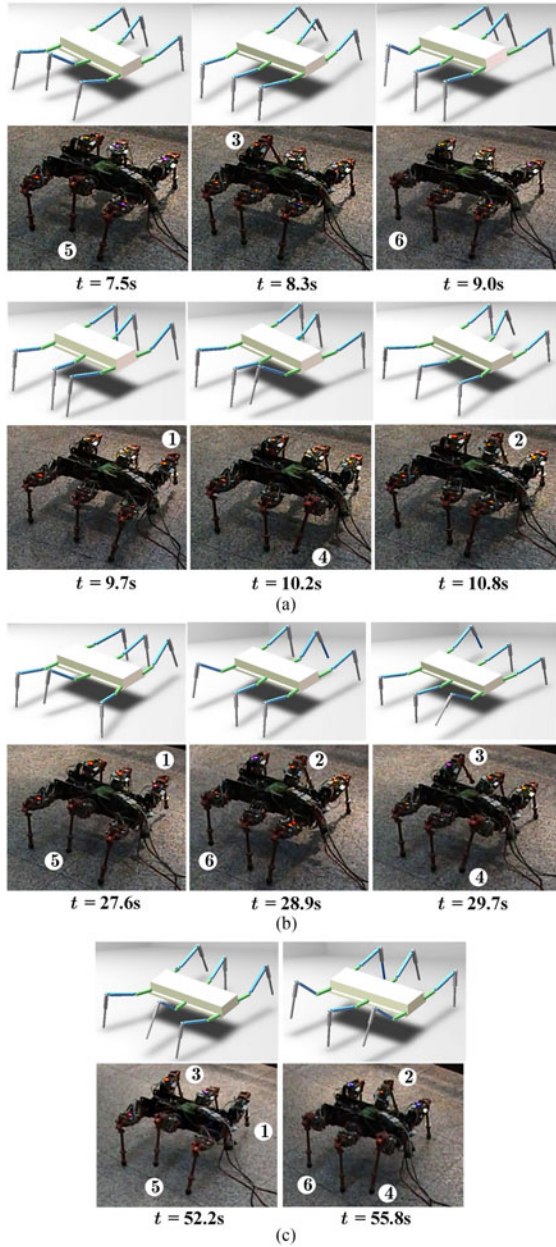


Fig. 11. Snapshots of the hexapod robot in gait transition with walking speed increasing from 0.033 to 0.138 m/s. (a), (b), and (c) are the key frames of the wave, tetrapod, and tripod gait, respectively (the comparative simulation results are placed on top). The robot is walking from left to right. Highlight numbers indicate the leg in swing phase.

The masses of the body and limbs is distributed according to [Table II](#) to ensure the simulation fidelity. The following parameters are used during simulation: $b = 4$, $e = 0.1$, $\omega_0 = 0.4\pi$, $\gamma = 1$ for the CPG network; the gait specifications are determined according to [Table 1](#) for individual gait and the transition rate is fixed at $\kappa = 1$ in (38).

The simulation results are shown in [Fig. 10](#). The robot starts at tripod gait and then transverse into tetrapod and wave gait subsequently (see [Fig. 10\(g\)](#) for the snapshot details). The commands for the tripod-tetrapod transition (entitled as Transition-I) and the tetrapod-wave transition (entitled as Transition-II) are,

respectively, triggered at $t = 20$ s and $t = 50$ s. The outputs of the CPG network is given in [Fig. 10\(a\)](#). The proposed network converges into the synchronous state with the desired phase lags as expected for all three gaits in regardless of the initial condition. According to (29) and (30), the phase-locked behavior of the network ensures that the robot swings its individual leg as the prescribed PS δ_i ($i = 1 \sim 6$) and the stable gait generation of the virtual hexapod model is guaranteed. Transition-I and II records the process that the network shifts the phase lags θ_i according to the smooth variation of the PS δ_i versus time [see [Fig. 10\(d\)](#)]. It can be observed that during the convergence period in Transition-I and II, the signals produced by CPG network exhibit sufficient smooth profile without jerky or sharp edge. Note that the profile of the DF β_{DF} [see [Fig. 10\(c\)](#)] has no influence on the evolution of the network since the resulting threshold σ only restricts the output x_i of each CPG unit.

[Fig. 10\(b\)](#) reveals the footprint of the walking pattern obtained via collecting the contact forces in simulation. For each scheduled gait, the robot exactly produces the stable walking pattern as expected. The leg may shorten or prolong its nominal swing/stance duration to cooperate with other legs such that the robot maintains moving forward without stop halfway during Transition-I and II. The original walking pattern is consequently disturbed by the transition commands. It is worth noting that the foot trajectory planning (34) and (35) are conducive to such disturbed transition pattern. Theoretically, the constant speed when foot contacts the ground could afford arbitrary entrancing (touchdown) and exiting (lift-off) the stance phase, which implies that the stance phase for each leg could be either enabled or interrupted when necessary. This advantage benefits the leg movement for the unconventional walking pattern especially in transition stage.

[Fig. 10\(e\)](#) and [\(f\)](#) records the body speed and displacement in simulation. The fluctuations of the simulated walking speed can be, respectively, observed in tripod ($-3.61\% \sim 5.50\%$), tetrapod ($-5.29\% \sim 4.523\%$), and wave gait ($-4.85\% \sim 3.94\%$) in contrast to the theoretical steady values. Foot slipping during simulation plays a key role in this phenomenon. Moreover, different gait types display specified “quasi-periodic” oscillation patterns of the body speed, which also results in the displacement difference between the simulated and theoretical values.

C. Walking Test on a Hexapod Robot

The walking test is implemented to examine the performance of the proposed strategy on the hexapod robot platform. The test begins at wave gait and transfers into tetrapod and tripod gait at $t = 20$ and 40 s, respectively. Meanwhile, the linear increment of the frequency of CPG network (from 0.4π to π) is implemented on the algorithm driving the robot to accelerate from 0.033 to 0.138 m/s. Furthermore, a comparative simulation that shares the identical algorithm with consistent input as exerted in the experiment is conducted to verify the validity of the modeling of hexapod robot in this paper.

The detailed snapshots as shown in [Fig. 11](#) have recorded the overall process of the walking test. Obviously, the typical gaits

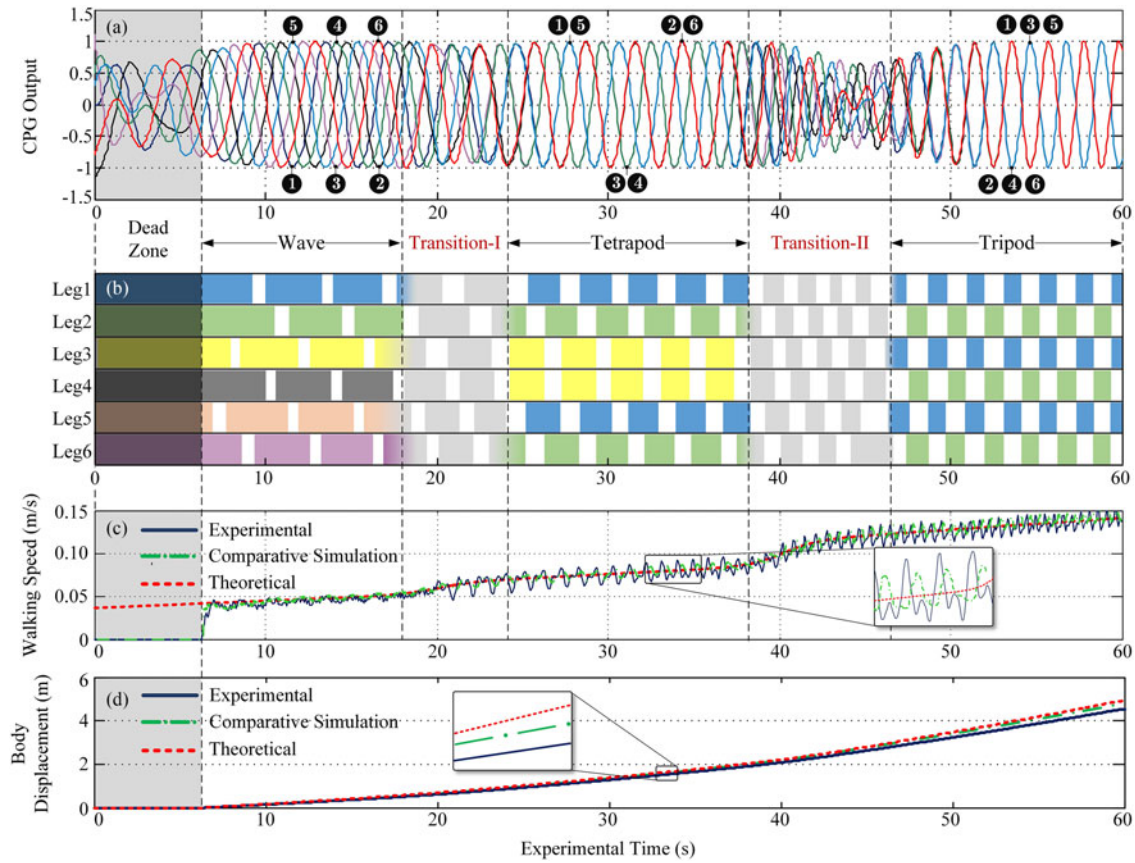


Fig. 12. Experiment results for the walking test on the hexapod robot in physical environment. (a) Outputs of the coupled CPG network. (b) Resulting footprint diagram of the gait cycle during gait transition. (c) Comparison of walking speed obtained in the theoretical computing, comparative simulation, and experiment. (d) Corresponding body displacement of the hexapod robot.

have been successfully generated on the hexapod robot during walking. The smooth gait transitions from wave to tetrapod, to tripod are also achieved via online adjusting the phase lags of the CPG network. Fig. 12(a) and (b) shows the associated CPG signals and the actual walking pattern obtained by analyzing the sampling data from the load cell mounted at the toe of each leg, which reveals that the proposed algorithm could generate stable typical gait and achieve smooth gait transition for the hexapod robot. Fig. 12(c) illustrates the walking speed variation during the experiment. It is clear to see that the speed fluctuation after the robot gets started. This observation could be explained as the toe slipping during the foot-ground interaction caused by the shortage of frictions. Such effect could be more obviously observed in tripod gait than in wave and tetrapod with relatively low speed. In addition, Fig. 12(d) shows the body displacement of the hexapod robot during the walking test. The theoretical value of the displacement is obtained by integrating the walking speed in (31) over time from the initial time instant $t = 6.5$ s.

The smooth gait transition is embodied in twofold. First, the original reference signals produced by the CPG network to allocate the swing/stance subphase is sufficient smooth whether in phase-locked period or not as observed in Figs. 10(a) and 12(a), avoiding any jerky, sharp or discontinuous signals. Therefore, the obtained swing/stance allocation ensures that individual legs travels in natural sequence. Second, the advantage of

the transition law (38) is to provide a sufficiently smooth variation for b_{DF} , thereby resulting in a smooth transition of the walking speed v according to (31). These two aspects provide a smooth gait transition process for the hexapod robot as observed in both simulation and experiment (see Figs. 10(e) and 12(c), respectively).

VI. CONCLUSION

In this paper, a novel CPG-based locomotion control method for hexapod walking robot has been proposed. The ring-shape CPG network consists of the VDP oscillator and is constructed by a simple diffusive coupling scheme, preserving a phase-locked behavior. The stability of the proposed network is proved using synchronization analysis. A corresponding locomotion control strategy is developed by employing the CPG network to govern the swing/stance phase for each leg. A foot trajectory planning is also presented to manipulate the leg movements during gait cycle, providing a constant AEP to overcome the unconventional leg touchdown and lift-off during gait transition. Using the proposed strategy with online phase lag adjustment of the CPG network, stable gait generation with smooth transition for hexapod robots is achieved. The effectiveness of the proposed locomotion control method has been verified by simulation and physical platform experiments. The hexapod robot could walk stably in tripod, tetrapod, and wave

gait while displaying smooth transitions among the generated gaits.

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Haitao Yu (S'11–M'15) received the B.S., M.S., and Ph.D. degrees in mechanical engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2007, 2009, and 2014, respectively.

He was a Visiting Scholar at Ryerson University, Toronto, ON, Canada from 2014 to 2015 before he joined HIT, where he is currently a Lecturer with the School of Mechatronics. His research interests include bioinspired robots and legged locomotion control.



Haibo Gao received the M.S. and Ph.D. degrees in mechanical engineering from Harbin Institute of Technology (HIT), Harbin, China, in 1995 and 2003, respectively.

He is currently a Professor and the Vice Dean of the School of Mechatronics, HIT, and the Vice Dean of the State Key Laboratory of Robotics and Systems, HIT. His research interests mainly focus on mobile robots, aerospace mechanisms, and control and legged robots.



Liang Ding (S'98–M'10) received the Ph.D. degree in mechanical engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2010.

He is currently a Professor with the School of Mechatronics, HIT. His research interests include mechanics (terramechanics), control and simulation for robots, and mechatronic systems, particularly for planetary exploration rovers and legged robots.



Mantian Li received the Ph.D. degree in mechanical engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2006.

He is currently an Associate Professor with the School of Mechatronics, HIT. His research interests include biomimetics and legged robots.



Zongquan Deng received M.S. degree in mechanical engineering from Harbin Institute of Technology (HIT), Harbin, China, in 1984.

In 1984, he joined the Department of Mechatronics, HIT, where he is currently the Vice President of HIT and the Director of the National Defense Key Laboratory of Aerospace Mechanisms and Control at HIT. His research interests include aerospace mechanisms and control and mobile robots.



Guangjun Liu (M'99–SM'08) received the B.E. degree from the University of Science and Technology of China, Hefei, China, in 1984, the M.E. degree from the Chinese Academy of Sciences, Institute of Automation, Shenyang, China, in 1987, and the Ph.D. degree from the University of Toronto, Toronto, ON, Canada, in 1996.

He is currently a Professor and the Canada Research Chair of Control Systems and Robotics with the Department of Aerospace Engineering, Ryerson University, Toronto. His research interests include control systems and robotics, particularly in modular and reconfigurable robots, mobile manipulators, and aircraft systems.